

Algorithmic Fairness 2.0

Empowering Users to Recognize, Interrupt, and Reshape Recommendation Systems through a 12- Days UX Research Diary Study

YouTube's algorithm serves **2 billion users daily**, optimising for watch time not wellbeing. It traps users in **self-reinforcing echo chambers**, silently narrowing their content diet every session without their awareness.

308

DATA POINTS

12

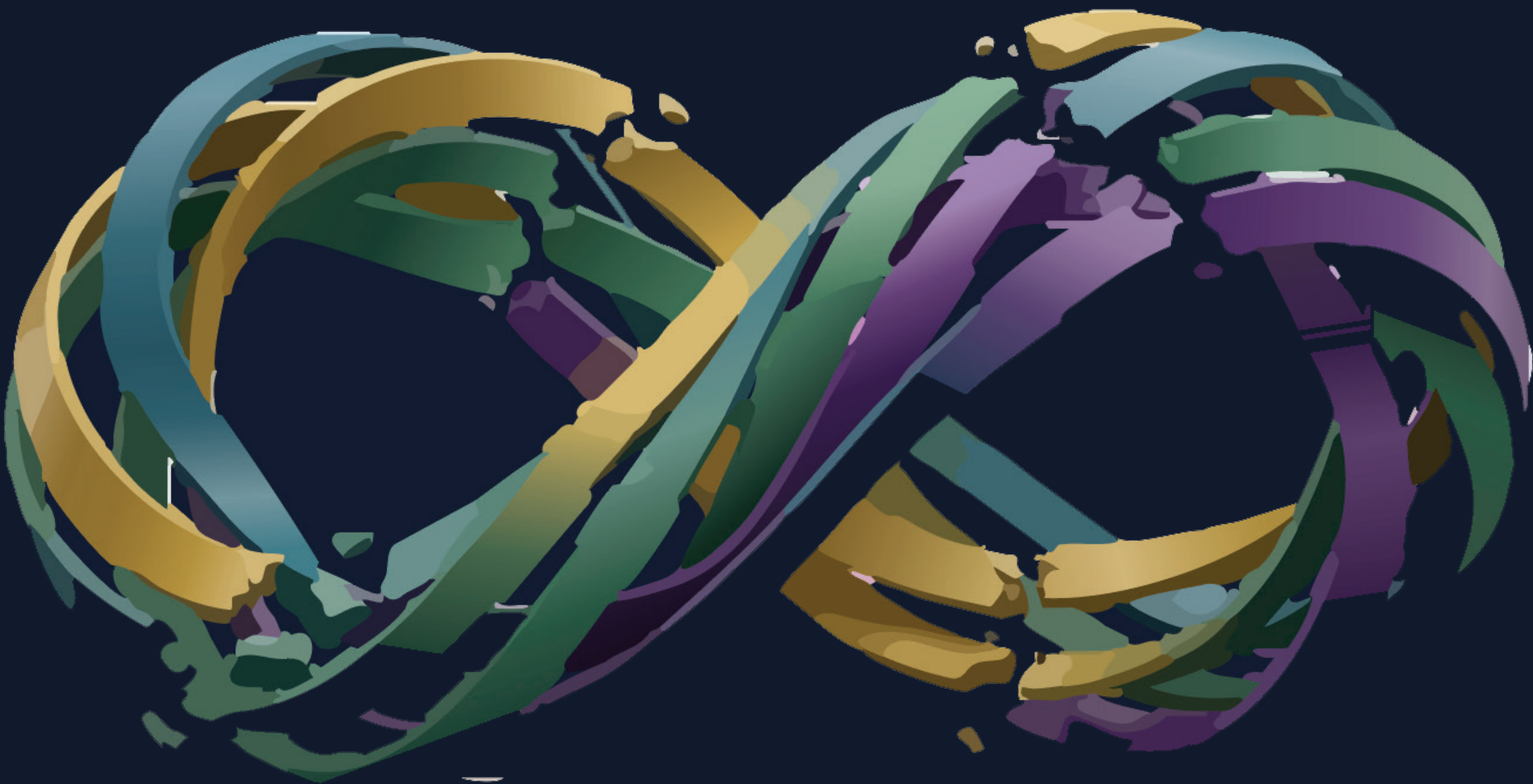
STUDY DAYS

+1.9

AVG. MOOD GAIN

72

DIARY ENTRIES
ANALYSED



Chef



IT Student



Musician



News Reader

4 - STEP EMPOWERMENT FRAMEWORK

I

Recognize the Bias

Observe your feed without changing behaviour
Days 1-3 | Goal: Awareness

II

Break the Loop

Disrupt usual viewing patterns intentionally
Days 4-6 | Goal: Disruption

III

Diversify Your Feed

Actively introduce new creators and content
Days 7-9 | Goal: Diversification

IV

Build Digital Awareness

Observe how YouTube shapes mood & autonomy
Days 10-12 | Goal: Cognitive Awareness

6 - CODED THEMES SCORED 0-9 PER SESSION



Echo Chamber

Repetitive, narrow content —
same creators every session



Algorithm Signal

Autoplay actively driving viewing
without user intent



Discovery

New creators & genres beyond
prior content history



Absence & Bias

Voices & genres systematically
excluded from the feed



Agency & Control

Intentional, self-directed session —
user steering



Emotional Impact

How YouTube affected mood, anxiety
& autonomy

| METHODOLOGY

TF-IDF
Theme Coding
LSA Sentiment Analysis
Co-occurrence
D3.js
Chart.js
HTML / CSS

| KEY FINDINGS

+1.9

3+ days

S I→IV

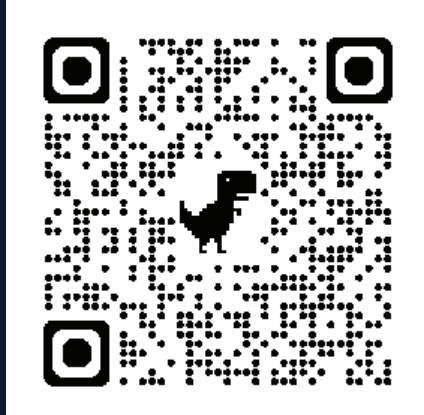
Section III

Measurable empowerment growth — average mood improved across all 4 participants on a 1–5 scale over 12 days

Sustained disruption required to overcome algorithm inertia and produce lasting feed diversification

Awareness always precedes control — Section I cognitive groundwork was essential to every subsequent gain

Highest-impact phase — mood and agency peaked simultaneously for all participants during Days 7–9



| SCAN HERE TO VIEW DASHBOARD

“12 Days, No tools, No settings changes. Just structured self reflection.”